

S P E C T R A V I D E O

U S E R S G R O U P -- Wgtn

Volume 4, Number 4 - MARCH/APRIL 1988
(JAN/FEB issue was replaced by meeting notice)

The second meeting for this year will be on Monday, March 21, 1988 at 7.30pm in the Staff room of Newlands Intermediate, Bracken Rd, Newlands. The following meeting will be held on the 18th April 1988, (ie as usual the 3rd Monday of the month.)

** PLEASE NOTE **

HAVE A THINK ABOUT WHERE YOU WANT THE CLUB TO GO THIS YEAR, AND COME ALONG AND HAVE YOUR SAY ... THIS NEWSLETTER IS ONLY PART OF THE CLUB'S ACTIVITIES. WITHOUT THE MEMBERS SUPPORT AND ENCOURAGEMENT AND NEW IDEAS THE CLUB WILL STAGNATE AND THERE WILL BE NOTHING TO WRITE ABOUT IN THIS NEWSLETTER. C'MON AND PARTICIPATE - COME TO THE NEXT MEETING!

----- Wellington SV Users Group

Committee Members

Chairman	-----	Jim Oliver 699-618
Secretary/Treasurer	-----	Ross Fletcher 788-454
Librarian	-----	Scotty Baxter 784-619

Newsletter Committee

Steve Hanson 843-495 (This is my LAST issue)
Phillip Daymond 845-974

Other areas of responsibility for committee members are:

- Program evaluation
- Jim -- Public Domain BASIC & Radio Access
- Ross -- Public Domain CP/M

DO YOU WANT TO INCREASE YOUR INVOLVEMENT WITH OUR CLUB ??
DO YOU WORK WELL AS PART OF A COMMITTEE ??

WOULD YOU LIKE TO HELP YOUR FELLOW CLUB MEMBERS,
WHILE LEARNING A FEW THINGS YOURSELF ABOUT THE S.V.?

HAVE I GOT THE JOB FOR **YOU** !!!!

AS **NEWSLETTER EDITOR** YOU WILL BE FULFILLING A
VITAL POSITION WITHIN THE CLUB. YOU WILL BE MAINTAINING THE ONLY
'TWO-WAY LINK' BETWEEN OUT-OF-TOWN MEMBERS, AND THE REST OF THE
CLUB. WITHOUT THIS LINK, OUR CLUB WOULD SURELY FAIL

APPLY TO JIM, OR CONTACT ME AT THE MARCH MEETING FOR DETAILS.
(ED: Steve Hanson)

WHAT IS INVOLVED ?

'Editor' does **NOT** mean that you have to write every article.

You have to co-opt, connive with, coerce, cajole, complement,
co-operate with, and sometimes 'con' other members into
providing meat for your bi-monthly sandwich of surprises.

Now don't get me wrong, .. there are many regular contributors
who provide the backbone of this magazine's content. In fact
without the regular provisions from Jim Oliver (particularly) Bob
Curwood, Don Rodgers, Darryl Alison, and a few others (unnamed
but not unappreciated) the last few issues would have been non-
existent.

What I mean is to get this (sometimes 20+ page) bulletin
compiled, printed, posted, and in member's hands takes some
organisation. You have got to ensure that articles are available
in time to be proof-read, (and if necessary re-typed), to test
programs, before they are included in the finished product. You
have to co-ordinate with the printer, and arrange to get the
labelled issues to the Post Office in Upper Willis Street two or
three posting days before a meeting (at least).

Above all else, you are responsible for the 'Public Image' of our
club. It is a condition of our bulk mailing permit that a copy of
each issue is sent to the National Library as a matter of 'Public
Record'. You must always keep at the back of your mind the club's
obligation to protect software author's rights. You also must be
aware that the newsletter must be the **IMPARTIAL**
mouthpiece of the Club. Every member has a right to have their
views/problems heard, and you must be wary that any unattributed
comments can be taken as being the 'voice of the club committee'.
This is a responsible position which should be taken as such. If
you or any of your contributors wish to 'stand up and be counted'
then you must ensure that such comments can only be taken for
what they are ... the personal view of the contributor. As editor
you must ensure that no anonymous contribution can be mistaken as
a comment of the committee, or club as a whole. But that's the
same for any club's magazine.

WHAT ARE THE QUALIFICATIONS REQUIRED

Anyone can handle the job, but some things make life easier:

As we are typing a fair amount of info, it can help HEAPS if you are familiar with Wordstar or any similar word-processor package.

Many contributions come in by disk, so it's advisable to use a disk based system (rather than a tape based processor like Just Write Jr), although being able to read a cassette tape is an additional advantage.

You must have access to a GOOD quality printer. - As you have seen with some of my earlier attempts, fuzzy or faint print does not reproduce well. You must also be able to photo-reduce the bulletin including any articles submitted to you as hard-copy. You must also co-ordinate with a printer, or arrange a straight photocopying job to produce the finished article, either process causing further loss of definition.

Other requirements are proofreading ability,.. a 'layout sense' helps,... a reasonable knowledge of the SV and it's related MSX brothers is a help in sorting out readers queries,.. and a preparedness to sit on the phone for a while.. 'cos you inevitably receive phonecalls from (distressed occasionally) SV owners looking for help with this or that. Provided you can co-opt people to help with the finer points (technical or programming-wise), these are no great trouble, and are what I found to be the most rewarding part of the job. Although I have personally found few problems, you must be prepared to accept criticism, as often as you accept praise for your work (wasn't it said somewhere "You can please some of the people...."?).

YOU MAY ASK WHY I'M GIVING UP

It's not because I am leaving the Club. I'll still be around to provide my 4+ years of experience with SV to those who want it.

Basically, I have been part of the Newsletter Committee since November 1985, and consider that it's about time someone else had a go. During this time I have been helped by a newsletter 'committee', but with work and other commitments, I am recently finding it difficult to keep in touch with those helpers at the time when the bulletin is due.

For an organised person, the workload involved in producing a Two-monthly newsletter does not need to be great, ... however I would be the last person to call myself "organised" ... in fact the antithesis applies.

When I first became involved, the newsletter was a way to learn as much as possible about the SV. It was a challenge which has with the passage of time (and heaps of it too) become somewhat of a chore. I hope that, by passing on the responsibility of co-ordinating the production, I will become a regular contributor myself. Here's Hoping...!

PRINT SCREEN ROUTINE

Do you ever wish that our Spectravideo computers had a 'prtscr' key like a certain other 16 bit machines ?

Alan MacPherson has kindly adapted an MSX routine from 'Introducing MSX Assembly Language and Machine Code' by Ian Sinclair which will quickly dump a 40 column screen to the printer.

Type in the following routine and run it. This will poke the required bytes into memory and also it should create a machine code file on disc *.

Whenever you hit that pretty useless 'PRINT' key you will get a fast print out of the 40 column screen on your printer.

PRTSCR.BAS

```
100 REM ENTER DEFUSRO=&HFCOE:X=USR(0)
110 REM TO PRINT THE 40 COLUMN SCREEN
120 REM EVERY TIME YOU HIT THE PRINT KEY
130 FOR I=&HFCOE TO &HFC5C: READ A$:POKE I,VAL("&H"+A$):
    S=S+VAL("&H"+A$):NEXT
140 'ONCE YOU HAVE CREATED THE M/C FILE
150 DATA F3,3E,C3,32,7B,FF,21,1C,FC,22
160 'YOU MAY USE IT BY SIMPLY TYPING
170 DATA 7C,FF,FB,C9,FE,45,C0,E5,C5,F5
180 'BLOAD "1:scrprt",R
190 DATA 3E,C9,32,7B,FF,FB,21,0,0,1,18
200 '
210 DATA 2B,CD,47,37,E5,E1,DB,84,CD,3C
220 '
230 DATA 3C,E6,7F,FE,20,38,3,CD,44,0,23
240 '
250 DATA 10,EA,3E,A,CD,44,0,3E,D,CD,44
260 '
270 DATA 0,6,2B,D,20,DB,F3,3E,C3,32,7B
280 '
290 DATA FF,F1,C1,E1,C9
300 '
310 IF S = 9844 THEN BSAVE "1:scrprt",&HFCOE,&HFC5E ELSE
    PRINT"DATA ENTRY INCORRECT, CHECK FOR TYPING ERRORS."
```

* If you use tape then simply omit the "1:" above and press the record and play keys when instructed to do so.

For those who are interested this routine is placed out of the way of any Basic programme. The one drawback for this luxury is the fact that this area is reset to zero after a warm boot. You may wish to alter the area of memory you use.

BENCH MARK TEST (SV VS. IBM)

By J Oliver

A long while ago I remember seeing an article in Bits & Bytes by Mr. Bridger on bench marks and SV. In that article it was shown that SV does pretty well against the 16 bit machines.

Recently I came across an MS-DOS program called Benchmark. In this is included a file called BENCH12.BAS which is to be used under BASICA on an IBM or similar machine.

Fortunately BENCH12 was easily converted to SV basic and I ran a series of tests to compare 'the one at work' with 'the one at home'.

The programme consists of 6 simple tests followed by one disc access test.

The times in seconds I measured were as follows:

		SV	IBM
TEST 1	Integer addition N1	61.6	52
TEST 2	Integer addition N2	60.7	42
TEST 3	Floating point calcs.	302.6	121
TEST 4	String concatenation	104.6	96
TEST 5	Data lookup loop	136.0	132
TEST 6	Empty FOR/NEXT loop	10.0	10
TEST 7	500 record random files (PUT then GET)	59.0	109

The original programme called for 1000 records in the disc test. SV single sided will only take about 570 odd so I chose 500.

The IBM has a clock speed of 4.77 MHz against SV at 3.58 MHz, so it really does pretty well as you can see. This test also shows the slowing down effect of double precision calculations. Note the very fast disc read / write speed.

Anyone interested is welcome to a copy of the converted program.

And now from the 'did you know department'.

During the process of converting BENCH12.BAS I wondered what it would be like to run it under CP/M using MBASIC. I gave up this course since it would mean converting all the LOCATE statements to that nasty ESC Y business. However I tried it under SUMBASIC just for fun and JOY!! SUMBASIC supports the LOCATE statement.

You knew that already ? Then why didn't you tell us ?

In fact I have found that SUMBASIC is much more friendly in other ways as well. It is not fussy about using capital letters as MBASIC is. Try it, I'm sure you will appreciate the differences.

INPUTTING TO SCREEN 1

(J Oliver)

We all know that to input to SCREEN 1 can be a bit tricky until you get the hang of it, but can you get the hang of this?

```
Try          10 SCREEN 1
              20 LINE INPUT D$
              30 PRINT D$
              50 GOTO 50
```

Simple? Now try inputting something (about 5 numbers at first) but don't hit the enter key, instead hit the backspace key a few times. Simple? Now RUN it again and input about 4 lines or more of rubbish then try the backspace key lots of times. Who said a vertical scroll in SCREEN 1 was difficult from BASIC?

Now RUN it again and hit CTRL plus any other key before <ENTER>. Simple?

```
Try adding this      40 D=VAL(D$):LOCATE 50,50:PRINT D
and RUN it again. The possibilities are endless!
```

PAINTING ON SCREEN 2

By J Oliver

When you paint onto screen 1 you must first ensure that the outline of the area to be painted is the same colour as the paint colour. Screen 2 allows you to paint in any colour to any colour border providing you specify the border colour. So the PAINT statement (in screen 2 only) becomes PAINT (X,Y),C,C2, where C is the paint colour and C2 is the border colour. To illustrate how this works type in the following program.

You will of course require a colour TV or monitor with good colour resolution since the differences between say light and mid green are quite small.

There's plenty of room for tarting up this game so see what you can do and then give it back to me for PD!

```
10 REM puzzle by J.A.Oliver WSV 480 REM setup random colours
   UG Dec 87                     490 SCREEN 0:COLOR 15,4,5:INPUT
20 CLEAR 1000:DIM C(25),C1(25):  "How many colours (max.14)"
   X=RND(-TIME)                  ;MC:PRINT:PRINT
30 ON STOP GOSUB 870:STOP ON    500 PRINT"Please wait . puzzle
40 CLS:PRINT "Do you want instr being calculated"
   uctions":IN$=INPUT$(1):IF IN 510 FOR T=0 TO 23:C(T)=INT(RND(
   STR("Yy",IN$)<> 0 THEN CLS:G  1)*MC+1)+1:T$=STR$(C(T)):T$=
   OSUB 660                      RIGHT$(T$,LEN(T$)-1):A$=A$+T$
50 GOSUB 490                     :NEXT T:T=0:RETURN
60 REM draw lines and paint are 520 FOR I=0 TO 23:J=INT(RND(1)*
   as                             (23+1))
70 SCREEN 2,1:COLOR1,15:CLS      530 SWAP C(I),C(J)
```

```

80 L1=154:L2=256:SC=1:'sprite=b 540 C1(I)=C(I)
   lack 550 NEXT
90 FOR Y=0 TO 192 STEP 32 560 RETURN
100 LINE(0,Y)-(256,Y),1 570 LINE (132,0)-(132,191),15:R
110 NEXT Y ETURN
120 FOR X=0 TO 256 STEP 32 580 REM check colors in LH area
130 LINE(X,0)-(X,192),1 s for equivalence to A$ in RH
140 NEXT X area
150 REM paint in area RH side 590 L1=16:L2=128
160 FOR X=L1 TO L2 STEP 32 600 FOR X=L1 TO L2 STEP 32
170 FOR Y=16 TO 191STEP 32 610 FOR Y=16 TO 191STEP 32
180 PAINT(X,Y),C(T),1 620 T=POINT(X,Y):T$=STR$(T):T$=
190 T=T+1:NEXT Y,X RIGHT$(T$,LEN(T$)-1):B$=B$+T$
200 L1=16:L2=128:T=0 630 NEXT Y,X
210 REM paint in work area 640 IF B$=A$ THEN PLAY "o5l64t2
220 GOSUB 520 55cdefgabc":FOR B=1 TO 15:
230 FOR X=L1 TO L2 STEP 32 COLOR ,,B:FOR D=1 TO 20:NEXT D,B
240 FOR Y=16 TO 191STEP 32 650 SC=15:B$="":RETURN
250 PAINT(X,Y),C1(T),1 660 PRINT " The object of PUZZL
260 T=T+1:NEXT Y,X E is to match the ";
270 GOSUB 570 670 PRINT
280 REM sprite routine 680 PRINT " coloured squares on
290 FOR T=1 TO 3:READ C$:D$=D$+ the left with the ";
   CHR$(VAL("&b"+C$)):NEXT T 690 PRINT
300 SPRITE$(1)=D$ 700 PRINT " squares on the right
310 E=10:F=16 (reference set). ";
320 PUT SPRITE 1,(E,F),SC,1 710 PRINT
330 REM cursor input routine 720 PRINT
340 J=STICK(0) 730 PRINT " You may 'rotate' th
350 IF J=1 THEN F=F-32:IF F<16T e colours by using ";
   HEN F=176 740 PRINT " the 'c' key. (ch
360 IF J=3 THEN E=E+32:IF E>106 ange)
   THEN E=10 750 PRINT
370 IF J=5 THEN F=F+32:IF F>192 760 PRINT
   THEN F=16 770 PRINT " You may swap two of
380 IF J=7 THEN E=E-32:IF E<10T squares by placing ";
   HEN E=106 780 PRINT " the cross over the
390 CC$=INKEY$:IF CC$=CHR$(27)T first square and ";
   HEN E1=E:F1=F:SC=15:OC=POINT 790 PRINT " pressing <ESC> then
   (E,F)'old colour move the cross to ";
400 IF CC$="s" OR CC$="S" THEN: 800 PRINT " the second square an
   PAINT (E1,F1),POINT(E,F),1:P d press 's' to swap";
   AINT (E,F),OC,1:SC=1's=swap 810 PRINT
   colours 820 PRINT
410 NC=POINT(E,F)+1:IF NC>MC+1 830 PRINT " If you think you ha
   OR NC<2 THEN NC=2:'new colo ve it correct you ";
   r 840 PRINT " press the 'w' key.
420 IF CC$="c" OR CC$="C" THEN ";
   PAINT(E,F),NC,1'rotate colou rs
430 IF CC$="w" OR CC$="W" THEN
   GOSUB 580
440 GOTO 320
450 DATA 00100100
460 DATA 00011000
470 DATA 00100100

```

DISC SYSGEN PROBLEMS

J Oliver

When I first started using discs I used to have a constant problem transferring the system tracks to a new disc. I used to have particular problems with Basic discs.

There are several methods available, some of which may be new to you. The following are the ones I have used along with some comments on the merits or otherwise of each method.

BASIC SYSTEMS :

The most obvious method here is to use SYSGEN.BAS supplied on your Basic System Disc. I find this program incredibly slow and it nearly always bombs halfway. I never use it now.

The next most common method is to use SYSGEN.COM off your CP/M System Disc. The method is to firstly boot up CP/M then type 'SYSGEN'. Now replace the CP/M disc in the 'A' drive with a basic disc containing complete system tracks. Then simply follow the prompts. You should, nine times out of ten, successfully transfer the system over to your 'non sysgened' disc.

Occasionally this does not work, normally due to bad formatting in the first place. The most reliable formatting is done using the CP/M COM file called 'FORMAT'. This verifies all sectors during formatting. The Basic file called 'SVFRMT' is fast but does not carry out verification.

If after reformatting for the umpteenth time my disc still refuses to accept the system tracks I next try 'SCOPY.COM' from Steve McNamee's customised Spectravideo CP/M system disc. (This is available from PD). SCOPY.COM will copy the whole disc to another disc OR JUST the system tracks. This method will often work when all others fail. The disc containing this customised version of CP/M also has programmes to format and verify your discs as well as the ability to read 12 different disc formats, fit another 30 odd K on your CP/M discs, show and alter function keys on screen and many other features.

Possibly the best method for Basic discs is to use L-FRMT.BAS from the Canadian PD. This is available on Public Domain Disc 20. This is a modified version of the basic 'FORMAT' written for Spectravideo by J.Suzuki and found on your Basic System Disc.

L-FRMT.BAS prompts you to format track 20 or check the disc (takes a while) and also write electronically on the disc the date of formatting, the disc name, and any program to be automatically run on boot up. The date can be read by several of the other programmes on Disc 20. The program then asks if you want to transfer the system to this disc. If you do, simply follow the prompts, if you don't then the program alters the File Allocation Table (FAT) so you may use the extra (normally wasted space) system tracks for normal programmes. You end up with 38 free tracks after this option is selected.

CP/M DISCS :

The only programmes to transfer your CP/M systems are the ones found on either of the two CP/M disc versions already mentioned above. If you have any discs which will not sysgen under CP/M then you have probably bought cheap discs. If they are not cheap then take them back and complain!

VERIFYING DISCS :

The fastest method is to run this Basic programme on your formatted disc:

```
10 CLEAR 1000
20 INPUT "Which drive ";D
30 FOR N=0 TO 39
40 A$=DSK1$(D,N,17)
50 PRINT "Track ";N;"verify OK"
60 NEXT
```

A more sophisticated method is to use 'VERIFY.BAS' from the Canadian P.D disc. This not only checks the formatting but will try to fix bad sectors, check for interleaved files, check for proper track 20 formatting etc.

The best way to check a CP/M disc is to use FBAD56.COM. This program will check and lock out any bad sectors on any disc.

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THATS SLIGHTLY INTERESTING DEPT.

by Jim Oliver

Did you know that you can't properly call up a FILES listing onto SCREEN 1 ? Well I can't with my computer. All I get is one file name then lockup !

When using the WASH or NSWP.BAS program off the Canadian PD disc you will find that if you try to copy a file to the other disc which already has a file on it of the same name as the one being copied you will get a message asking if you want to delete the first one.

There are very few books or magazines around which relate directly to the SV range of computers. However recently I have been reading COMPUTE!. This particular magazine specialises in a selection of well known computers, the IBM PC among them.

Most of the programmes featured are translated to suit the computers supported. You are probably aware that the BASIC interpreter used by IBM was written by Microsoft, the same company that wrote SV and MSX BASIC, consequently there is a great similarity between the IBM listings and an SV listing.

I have recently converted some of these listings from IBM to SV including the IBM proof reader. This is an excellent system for typing in programmes and will eliminate ALL the typing errors resulting in a programme which runs FIRST time! (A revelation for me) This particular proof reader requires all the spaces etc to be correct before giving the right checksum. It however supports the full screen editor, the checksum being calculated and shown for each line of code. You then compare this to the magazine listing and either accept it or not. (The other proof readers in our PD will not accept the line until it is typed correctly)

Once the programme is typed in correctly it may then be edited to suit SV with your favourite editor (I use Wordstar for the first run through) and very quickly you will have a new game or whatever for your SV.

The IBM uses colours on it's text screen similar to our SCREEN 1 (very handy) and since most of the games use the IBM text screen the conversion to SV is simple but you have to be happy with just two colours. The GET and PUT graphics which IBM use to simulate sprites (since big blue does not support them) are very close to the SV use of them and I have run two programmes where no changes at all were needed.

In many cases the only change needed is to reverse the order of the LOCATE positions. eg IBM - LOCATE 10,15 ; SV - LOCATE 15,10.

If enough interest is shown in this magazine as a source of new programmes or ideas I will put together some notes on converting IBM to SV.

Incidentally, I know it sounds like I have shares in COMPUTE! (not true) but it is an excellent all round computer magazine and cut above the rest.

SPIN OFF PATCH

by Jim Oliver

I have now had a chance to look into the question of the spin off patch. The following is the patch which is run on your basic disc to alter the delay before switching off the drives.

```
20 REM This spin off patch will cause
30 REM a 6 second delay and stop those Disc I/O errors.
40 CLEAR 2000
50 A$=OSKI$(1,0,12)
60 MID$(A$,20,2)=CHR$(20)+CHR$(1)
70 FIELD #0,128 AS B$,127 AS D$
80 LSET B$=LEFT$(A$,128)
90 LSET D$=RIGHT$(A$,127)
100 DSKO$ 1,0,12
```

The original programme used in line 60 CHR\$(20)+CHR\$(0) and will switch off the drives almost instantly. This is too fast for some programmes (BASIC.IT for example) and can result in bad files.

An original Basic Master disc has CHR\$(8)+CHR\$(7) but a little experimenting shows that you can use a wide variety of values.

```
CHR$(10)+CHR$(1) gives about 7 seconds delay
CHR$(50)+CHR$(1) " " 10 " "
CHR$(50)+CHR$(2) " " 12 " "
```

So now you can alter the delay to suit your particular requirements. Remember that it is the second of the two bytes which is the most important.

See also SVUG Wgtn. Newsletter No 6, p 10.

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THATS SLIGHTLY INTERESTING DEPT. PART II
by Jim Oliver

This question has been asked before but never answered. Why is it not possible to VPOKE any of the non existant characters, i.e any with an ASCII value between 125 and 159 and any > 215 ? In theory it should be possible.

To find the size in bytes of any BASIC program in memory use this

```
PRINTHEX$((PEEK(&HF7EE)+256*PEEK(&HF7EF))-(PEEK(&HF54A)+256*PEEK(&HF54B)))
```

This subtracts the memory position start of any BASIC program from the start of the variable table and gives a pretty good indicator of any program size. The variable table starts around 2 or so bytes after the end of any basic program. If you save your program as a normal tokenised save then it's size will be very close to that given by the expression above.

I thought it might be a good idea to sort some of my disc directories into alphabetical order using the DISK.UTL from Bryan Fisk.

After I had done a few I decided to look at the FILES.

SHOCK... HORROR... I had lost some of my precious files, at least they weren't showing on the FILES printout. I then noticed that ALL the discs had a maximum of only 15 files, even though they started out at around 26 or so. However, being a wily old dog I had taken the precaution to save the directory as a backup, again using DISK.UTL. So I simply ran the 'load for backup' and viola! there they all were again. PHEW.

I eventually traced the problem.

Line 3550 is 'FOR X=1 TO 15'. It needs to be changed to 'FOR X=1 TO 16. I won't tell you why, you work it out!

(Hint: directory sectors have 16 files each)

P.S. You may have noticed that even after the files are sorted they only show up in true ASCII order on the 40 column screen. This is due to the cunning way Bryan has arranged it.

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At the request of Roger Higginson, I repeat an article on converting an SV 801 to run double sided disk drives.

SV-801 DISK CONTROLLER

During the investigation that led to a means of modifying the SV-801 Disk Controller for double-sided operation (see separate item elsewhere), I discovered that my own module had been modified by SV before delivery in a way that was not consistent with good circuit design. This modification results in two TTL output chips passing for more current than they are rated for, particularly if you are already using two disk drives. It may exist in other SV-801 modules and should be removed, irrespective of whether you intend to modify the module for double-sided operation.

To check your SV-801 module, there is no alternative but to disassemble it and remove the printed circuit board. On the end of the board near the drive connectors, identify IC7 and IC9 (either 7437 or 7438 devices - mine were 7438s). Turn the board over and check whether or not there are any 150 ohm resistors soldered from pins 3, 6, 8, 11 of IC7 and IC9 to the +5 Volt line (pin 14 of IC7 and IC9). If there are, remove them, being careful not to leave behind any solder bridges between adjacent IC pins.

After making a final check that all is in order, you can now reassemble the module and put your system back together. No special testing is required to confirm proper operation.

For the technically-minded, the 8 resistors referred to were connected to the output pins of the four line-driver gates in each chip, presumably to act as "pull-up" resistors for these particular disk drive control lines. It just happens that these lines are already terminated by 150 ohm resistors in each disk drive anyway and the extra components are thus unnecessary, as well as increasing the current flowing in the output stage of each gate. When two drives are connected, the line current with the resistors installed is well above the maximum recommended for the particular devices - hence the need to remove the resistors.

John Bridger

USING DOUBLE-SIDED DISK DRIVES

Until the arrival on the scene of the SV-605B Expander, with its 2 40-track double-sided disk drives, the SV machines have been restricted to using single-sided disks only. Even now, the version of Disk Basic for the SV-605B is still restricted to single-sided operation, although a revision is said to be in the pipeline. CP/M is, however, in a different situation as the operating system itself can be modified to use any combination of disk-drive types - single- or double-sided, 40- or 80-track. CP/M2.22 is configured for 2 single-sided 40-track drives, whilst CP/M2.23 is set up for 2 double-sided 40-track drives. Several systems have been operated successfully with one single-sided 40-track drive and one 80-track drive. Although 2 80-track drives could be used, it is probably wise to retain one 40-track unit in the system to retain compatibility with standard SV format - particularly if you want (or need) to be able to swap data with other SV users. For the meantime at least, Disk Basic will only operate on 40-track drives.

If you have an SV-601 Expander and an SV-801 Disk Controller or an SV605 Expander and if you have been thinking of adding extra disk capacity beyond the original drive, it is now worth considering getting a bare double-sided unit from the likes of Mitsubishi or Shugart, rather than another SV-902 (if you can get one) or the single-sided drive offered to upgrade the SV-605 to an SV-605A. This is because some inspired thinking by Ross Wakelin and some joint detective work on circuit design and board layout have resulted in a modification for the Disk Controllers so they can operate successfully with double-sided drives. The modification involves some fine soldering work on the controller printed circuit board and, for the SV-801, rearrangement of one wire on the ribbon cable to the disk drives. It will take 2 to 3 hours to modify an SV-801 and its cabling and about half that for the SV-605.

The modifications are not "magic", in that they will not turn an SV-902 drive (or the original one in an SV-605) into a double-sided drive - neither of these units has the second read/write head required for that mode of operation.

Before starting on any modification work at all, it is worthwhile insurance to make a duplicate copy of one of your present disks for use in proving your work afterwards. If anything untoward happens, at least you haven't lost any precious data!

SV-801 Modification

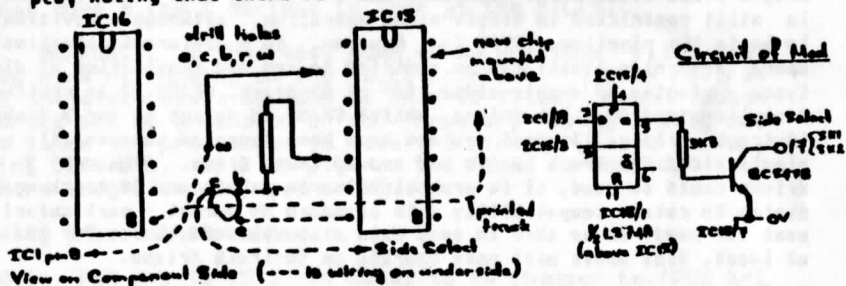
You will need the following new parts:-

- 1 off 74LS74A Dual flip-flop (no other 74..74 version will do)
- 1 off BC547B transistor (a BC548 or BC549 might do, but they have lower voltage ratings)
- 1 off 3.3 kohn 1/4 watt resistor
- 1 short length (300 mm) small gauge insulated wire (wire-wrap wire is ideal)

Along with the other tools you'll need, you'll also want a fine twist drill bit (about 0.8 mm). If you don't have one, I'm told that a steel dress-making pin with its head cut off (point end in the chuck) works quite well, even if it does take a little longer!

The procedure is as follows:-

- (a) disassemble the SY-801 module and identify IC15 in the top right corner of the pcb (near the drive connectors).
- (b) referring to the sketch below, locate and drill five holes in the pcb, noting that there is one under-side track to avoid.



- (c) install the transistor in the holes indicated, being careful to get its orientation correct. Bend its emitter lead flat below the board and solder it to IC15 pin 7. Cut both remaining leads to about 3-4 mm below the board and bend the base lead flat against the pcb, alongside the hole for the resistor.
- (d) carefully bend pins 2,5,6,8,9,10,11,12,13 of the new 74LS74A outwards, until they lie horizontal. Mount the new chip above IC15 (another 74LS74A) and solder pins 7,14,1,3,4 (in that order) to the corresponding pins of IC15.
- (e) solder the resistor between pin 5 of the new 74LS74A and the base of the transistor, bending its leads so it fits neatly.
- (f) on the underside of the pcb, solder one end of a length of wire to pin 8 of the FDI793 Floppy Disk Controller chip, IC1. Pass the free end through the remaining hole drilled earlier, routing it along the pcb centreline. Solder the other end to pin 2 of the new 74LS74A.
- (g) on the underside of the pcb, bend the transistor's collector lead flat against a bare area of board. Solder one end of a length of wire to the collector lead and route the wire along the centreline of the pcb to pin 17 of both disc drive connectors. Solder it in place, being careful not to leave any solder bridges to adjacent areas of copper (there isn't much room to spare). Pin 17 becomes the Side Select output from the controller to the drives.
- (h) mount the pcb in the bottom half of its case temporarily and mark where the case must be notched near the drive connectors so the Side Select wire will fit neatly. Remove the board and cut the notch (use a file).
- (i) make a final check of your work to ensure it's neat, tidy, and CORRECT. Then reassemble the module.

For the sake of consistency, it's probably a good idea to modify the ribbon cable connections to your present drive, at the drive end. After extracting the drive from its case, identify wire 17 in the cable from the controller (it probably goes to pin 14 on the drive pcb edge connector). Unsolder (or cut) it and reconnect it, extended if necessary, to pin 32 of the drive pcb edge connector. Check the cable continuity from the controller end right through to the drive connector and, if satisfactory, reinstall the drive in its case. You can now test your modified system (see later).

SV-605 Modification

You will need the following new parts:-

- 1 off BC547B transistor (a BC548 or BC549 might do, but they have lower voltage ratings)
- 1 off 3.3 kohm 1/4 watt resistor

The procedure is as follows:-

- (a) remove the outer cover of the Expander and any disk drives already installed.
- (b) check that the Floppy Disk Controller chip installed on the pcb (the 40-pin device) is an MB8877A. If so, you can proceed. If not, you have one of the very early SV-605s, which have a different pcb and to which an extra 74LS74A has to be added. Unfortunately for you, the modification details for this interim SV-605 pcb have not been worked out - sorry about that.
- (c) looking from the front of the SV-605, identify the positions marked on the pcb for transistor T3 and resistor R20 in the rear left corner of the main pcb. (The expansion slot pcb obstructs the under-side of the main pcb in this area, so the new components have to be soldered in place from the top of the board). Carefully solder the new parts into their correct locations as T3 and R20, ensuring that the transistor is installed the right way around.
- (d) check that your work is correct (T3's collector should connect to pin 32 on the drive pcb edge connector) and then reinstall the disk drive(s). Refit the Expander's cover.

Testing

You can now reassemble and test your modified system. Try both read and write exercises and fill up the disk, keeping track of progress by monitoring the disk directory. If all proceeds well, you're away with a disk controller that can cope with double-sided disk drives! If, however, the drive makes odd noises, you've probably left a solder-bridge somewhere in the controller. Check your work on the controller pcb carefully before you go any further.

Good luck with your double-sided future!

John Bridson